S4 — The effective action for composite operators of Cornwall, Jackiw and Tomboulis, applied to the two spacetimes models

Supplement to both manuscripts of PROJECT spacetimes: the method, independent checks of the earlier results, and what the method predicts beyond them — with commented source code and the provenance of every number

Prepared for Patrick Lee Nash, Ph.D.

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0. Purpose, and summary of the results

The author asked that the method of J. M. Cornwall, R. Jackiw and E. Tomboulis (Phys. Rev. D **10**, 2428 (1974); "CJT" below) be understood, be used as a check of the computations already made for the two models, and be used to obtain every further prediction that it gives. This supplement does the three things in that order. Section 1 explains the method for a reader who has not met it, and tests it on a model that can be solved exactly. Section 2 writes down the CJT action of the two models. Sections 3 to 7 apply it. Every number is produced by the command `spacetimes cjt` of the Rust program (source files `cjt_toy.rs`, `cjt_pairing.rs`, `cjt_normal.rs`, `cjt_vacuum.rs`, `cjt_cmd.rs`, and `symmetry.rs`); Sec. 9 says which part produces which number.

The symbols are those of "dirac matrices" and of Supplement S1 (cited as S1): the fermion field of mass m with $\nu = 8$ internal states per momentum; the Vorton-Scalar with vacuum value v_{VS}, radial mode s_{VS} of mass m_s and massless phase mode ϑ_{VS}; the coupling $g_{VS} = m/v_{VS}$, $\alpha_{VS} = g_{VS}^2/4\pi$. Unless stated otherwise the results hold for both spacetimes, because the matrices of 7+1 and of 4+4 give the same pairing kernels and the same internal symmetry in the sector used (S1, Secs. 2.1 and 4.2); where the two differ, it is said.

Checks of earlier results (all passed):

Earlier result	Check made here	Agreement
gap function, fixed-point iteration (S1 Sec. 5.2)	Newton's method on the CJT potential, different starting function	1.1×10^{-11}
gaps stored in `data/gap_scan.csv`	recomputed	7.9×10^{-9}
the solution is the state of lowest energy	minimum of the potential along rescaled gap functions; all second variations positive	minimum at the solution within 5.1×10^{-7}
critical temperature from the linearized gap equation	sign change of the second variation of the potential	2.6×10^{-12}
critical coupling of the bound pair, 1.6798 (S1 Eq. (5.4), `dimer.rs`)	threshold of the gap equation in vacuum, in momentum space	1.67982
Ginzburg–Landau coefficients a, b, K (S1 Secs. 6.2, 7)	expansion of the potential; pair susceptibility	within 1.0 per cent
quartic invariant and ground states (S1 Eq. (6.2), Sec. 6.3)	potential on the eight-dimensional space of the order parameter, from the matrices	confirmed, both signatures

One earlier statement was found to be wrong in the course of these checks and has been

corrected: the degeneracy of the hidden-axis singlet and the triplet in 4+4 is not accidental but follows from a hidden symmetry (S1, Sec. 2.1; error log E26).

Further predictions (each derived and qualified in the section named):

1. *Thermodynamics of the paired state* (Sec. 4.1): gap, free energy, entropy and heat capacity against temperature; $\Delta(0)/T_c = 1.7668$, jump of the heat capacity 1.433 times the normal value.
2. *The space of the order parameter* (Sec. 4.2): two fourfold gaps for a general state; four gapless directions around the ground state in both signatures; a stationary "half-paired" state with half the condensation energy.
3. *The fermion mass in matter* (Sec. 5): it falls with density and vanishes above a density at which the Vorton-Scalar loses its expectation value; matter is self-bound if $C = g_{VS}^2 m^2/m_s^2 > 3.7011$ (5.9218 with the energy of the negative-energy states).
4. *Finite temperature* (Sec. 6): the vacuum value and the fermion mass melt at a temperature between 0.4 and 7 fermion masses for the couplings examined.
5. *Vacuum* (Sec. 7): a pair condensate without a medium requires $\alpha_{VS} > \alpha_c$, with $\alpha_c \to \pi/2$ for a large cutoff.

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes"
cargo test --release --manifest-path code/spacetimes-rs/Cargo.toml   # 63 unit tests (23 for S4)
S="cargo run --release --manifest-path code/spacetimes-rs/Cargo.toml --"
$S symmetry        # data/reduced_symmetry.csv
$S cjt             # all parts (about one minute); data/cjt_*.csv
$S cjt pairing     # one part: toy pairing thermo threshold landscape normal thermal vacuum
python3 code/scripts/make_figures.py cjt_toy cjt_thermo cjt_normal cjt_thermal cjt_vacuum
python3 code/scripts/make_tables.py
```

The command `cjt` ends with an error status if any of its built-in requirements fails; each part prints "all requirements met" otherwise.

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1. The method

1.1 Why a second variable is needed

The usual effective action $\Gamma[\phi]$ of a quantum field theory is a functional of the expectation value $\phi(x) = \langle\Phi(x)\rangle$ of the field; its stationary points are the possible states, and for constant ϕ it reduces to the effective potential, whose minimum is the vacuum. It is the right tool when the quantity that acquires an expectation value is a *field*.

In a superfluid of fermions it is not. A fermion field has no expectation value; what becomes non-zero is the expectation value of a *product of two fields*, $\langle\psi(x)\psi(y)\rangle$, a "composite operator". The same is true whenever a symmetry is broken dynamically, by a condensate of pairs, and not by an elementary scalar. CJT constructed the effective action for this situation: a functional $\Gamma[\phi, G]$ of two variables, the expectation value $\phi(x)$ and the full propagator $G(x, y)$, the connected expectation value of the product of two fields. An expectation value $\langle\psi\psi\rangle$ is simply one component of the propagator of the fermions.

1.2 Definition

We use the Euclidean (imaginary-time) form throughout, because all applications below are to equilibrium; $S[\Phi]$ is the Euclidean action. For one real scalar field, add to the action a source $J(x)$ for

the field and a second source $K(x,y)$ for the product of two fields:

$$Z[J,K] = e^{W[J,K]} = \int \mathcal{D}\Phi \ \exp\Big\{ -S[\Phi] + \int_x J\Phi + \frac{1}{2}\int_x\int_y \Phi(x)K(x,y)\Phi(y)\Big\}. \tag{1.1}$$

The derivatives of W define the two variables:

$$\frac{\delta W}{\delta J(x)} = \phi(x), \qquad \frac{\delta W}{\delta K(x,y)} = \frac{1}{2}\big[\phi(x)\phi(y) + G(x,y)\big]. \tag{1.2}$$

The CJT effective action is the Legendre transform of W with respect to *both* sources,

$$\Gamma[\phi,G] = -W[J,K] + \int_x J\phi + \frac{1}{2}\int_x\int_y \big[\phi(x)\phi(y) + G(x,y)\big]K(x,y), \tag{1.3}$$

and it follows at once that

$$\frac{\delta\Gamma}{\delta\phi(x)} = J(x) + \int_y K(x,y)\phi(y), \qquad \frac{\delta\Gamma}{\delta G(x,y)} = \frac{1}{2}K(x,y). \tag{1.4}$$

The physical theory has no sources. **The physical ϕ and G are therefore the stationary point of Γ in both variables.**

1.3 The result of CJT

CJT showed that Γ has a remarkably simple structure:

$$\boxed{\Gamma[\phi,G] = S[\phi] + \frac{1}{2}\mathrm{Tr}\ln G^{-1} + \frac{1}{2}\mathrm{Tr}\big[D^{-1}(\phi)\,G - 1\big] + \Gamma_2[\phi,G]} \tag{1.5}$$

(the central result of Sec. II of their paper, written here in Euclidean form). The first term is the classical action. $D^{-1}(\phi;x,y) = \delta^2 S/\delta\phi(x)\delta\phi(y)$ is the inverse of the propagator of small fluctuations around the classical field ϕ. The traces are over the space-time arguments (and over internal indices, if any). The last term,

$$\Gamma_2[\phi,G] = -\big(\text{sum of all two-particle-irreducible vacuum diagrams}\big), \tag{1.6}$$

is computed with the vertices that the action $S[\phi+\varphi]$ has for the fluctuation φ (those of third and higher order in φ), and — this is the essential point — with the **full propagator G on every line**. A diagram is *two-particle irreducible* (2PI) if it does not fall into two pieces when any two of its lines are cut. Diagrams that do fall apart would count twice what the full propagator on the lines already contains.

Why this form. For a free field $\Gamma_2 = 0$, and the first three terms of (1.5) are the exact result of the Gaussian integral (1.1). The interaction adds Γ_2; the restriction to 2PI diagrams is what remains of the perturbation series when all self-energy insertions have been absorbed into the lines. The proof is in Sec. II of the CJT paper.

1.4 The stationarity conditions

Varying (1.5) with respect to G gives

$$G^{-1} = D^{-1}(\phi) + \Sigma[\phi,G], \qquad \Sigma \equiv 2\,\frac{\delta\Gamma_2}{\delta G}. \tag{1.7}$$

This is the exact Schwinger–Dyson equation for the propagator: the derivative of a 2PI vacuum diagram with respect to one of its lines is a self-energy diagram, and the self-energy is expressed

through the full propagator itself. Equation (1.7) is called the **gap equation**, because in a superconductor its off-diagonal part determines the energy gap. Varying (1.5) with respect to ϕ gives the field equation of the expectation value, with all fluctuations included.

For fields that are constant in space and time, Γ is proportional to the volume of space-time, and

$$V(\phi, G) = \frac{\Gamma[\phi, G]}{\text{volume} \times \text{time}} \tag{1.8}$$

is the **effective potential for composite operators**. At temperature T the imaginary time has the period $1/T$, and the value of V at its stationary point is the thermodynamic potential per unit volume (minus the pressure).

1.5 Fermions, and pairs

For a fermion field the same construction gives (1.5) with two changes that follow from the anticommuting integration variables: the two trace terms change sign, and there is no factor $\frac{1}{2}$ because ψ and $\bar{\psi}$ are independent,

$$\Gamma[S] = -\mathrm{Tr}\ln S^{-1} - \mathrm{Tr}\big[S_0^{-1}S - 1\big] + \Gamma_2[S]. \tag{1.9}$$

A propagator of this kind contains $\langle\psi\bar{\psi}\rangle$ only. To allow pair correlations $\langle\psi\psi\rangle$, one combines the field and its conjugate into one doubled field $\Psi = (\psi, \psi^{\dagger T})$ (Nambu, Gor'kov). Its propagator $\mathcal{S}$ has four blocks; the off-diagonal blocks are the pair correlations. The components of Ψ are not independent, which restores the factors $\frac{1}{2}$:

$$\Gamma[\mathcal{S}] = -\frac{1}{2}\mathrm{Tr}\ln \mathcal{S}^{-1} - \frac{1}{2}\mathrm{Tr}\big[\mathcal{S}_0^{-1}\mathcal{S} - 1\big] + \Gamma_2[\mathcal{S}]. \tag{1.10}$$

This is Eq. (5.1) of S1.

1.6 Approximations, and what they do and do not guarantee

Γ_2 is an infinite sum; an approximation keeps some of its diagrams. The natural ordering is by the number of loops. The diagrams with two loops give the approximations known by older names: for a four-boson interaction the *Hartree* approximation; for fermions exchanging a boson the *Hartree–Fock* approximation, and, with pair correlations, the theory of Bardeen, Cooper and Schrieffer. Even the simplest truncation is not perturbation theory: because the lines carry the full propagator, the gap equation (1.7) sums an infinite class of perturbative diagrams self-consistently.

Guaranteed by any truncation of Γ_2: the thermodynamic quantities derived from the stationary value of V are consistent with each other (the density is the derivative of the pressure, and so on), and conservation laws of symmetries that the truncation respects hold (Baym). The value of V at the stationary point has an error of second order in the error of the propagator.

Not guaranteed: the propagator that solves a truncated gap equation need not satisfy the identities that the exact propagator satisfies. The best-known example appears in Sec. 6: in the Hartree approximation the mode that must be massless by Goldstone's theorem acquires a small mass.

1.7 A test in zero dimensions

Everything above can be tested on a model in which nothing is approximated: a "field theory" of a single real variable, with the action

$$S(\varphi) = \frac{1}{2}\varphi^2 + \frac{\lambda}{24}\varphi^4.$$

Its partition function $Z = \int d\varphi\, e^{-S}/\sqrt{2\pi}$ and its "propagator" $G = \langle\varphi^2\rangle$ are ordinary integrals. The CJT action (1.5) is a function of one number,

$$\Gamma(G) = \frac{1}{2}\ln\frac{1}{G} + \frac{1}{2}(G-1) + \Gamma_2(G), \qquad \Gamma_2 = \frac{\lambda G^2}{8} - \frac{\lambda^2 G^4}{48} + \frac{\lambda^3 G^6}{48} - \frac{5\lambda^4 G^8}{128} + \dots, \tag{1.11}$$

whose terms are the 2PI diagrams with two, three, four and five loops. The gap equation (1.7) is $1/G = 1 + \Sigma$ with $\Sigma = \lambda G/2 - \lambda^2 G^3/6 + \lambda^3 G^5/4 - 5\lambda^4 G^7/8 + \dots$, and the value of Γ at its solution should equal $-\ln Z$.

Origin of the coefficients. From the Gaussian moments $\langle\varphi^{2k}\rangle_0 = (2k-1)!!$ the perturbation series of the exact propagator is $G = 1 - \lambda/2 + 2\lambda^2/3 - 11\lambda^3/8 + 34\lambda^4/9 - \dots$; inverting it and expressing $1/G - 1$ through G itself gives the series for Σ. The unit test `cjt_toy::tests::two_particle_irreducible_coefficients` verifies the four coefficients against the exact integral.

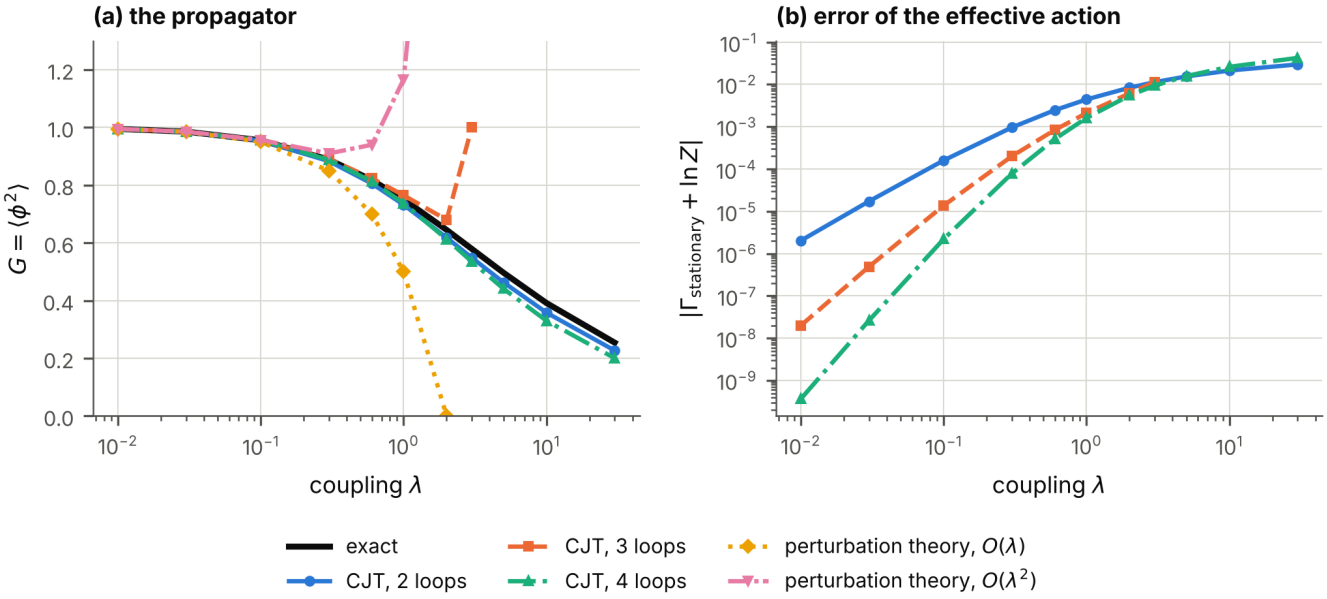


Figure 1: The CJT method in zero dimensions. (a) The exact propagator, the solutions of the gap equation with Γ_2 truncated at two, three and four loops, and ordinary perturbation theory. (b) Difference between the effective action at the stationary point and the exact $-\ln Z$. Data: `data/cjt_toy.csv`.

λ	exact	2 loops	3 loops	4 loops	5 loops	pert. $O(\lambda)$	$O(\lambda^2)$	$O(\lambda^3)$
0.1	0.9556	0.9545	0.9557	0.9555	0.9556	0.950	0.957	0.955
0.3	0.8887	0.8830	0.8905	0.8878	0.8894	0.850	0.910	0.873
0.6	0.8176	0.8054	0.8240	0.8125	0.8242	0.700	0.940	0.643
1	0.7505	0.7321	0.7646	0.7373	0.9776	0.500	1.167	-0.208
2	0.6449	0.6180	0.6801	0.6127	—	0.000	2.667	-8.333
3	0.5792	0.5486	1.0000	0.5351	—	-0.500	5.500	-31.625
5	0.4966	0.4633	—	0.4411	—	-1.500	15.167	-156.708
10	0.3915	0.3583	—	0.3300	—	-4.000	62.667	-1312.333

Three lessons, which carry over to field theory:

1. **The two-loop truncation is uniformly good.** It stays within 11 per cent of the exact propagator for all couplings from 0.01 to 30, where the perturbation series is useless beyond $\lambda \approx 0.5$. (Its solution, $G = (\sqrt{1+2\lambda}-1)/\lambda$, has the correct qualitative behaviour for $\lambda \to \infty$.)
2. **More loops help only at weak coupling.** The series (1.11) is asymptotic, like the perturbation series. At $\lambda = 1$ the error of the effective action falls from 4.4×10^{-3} (two loops) to 1.6×10^{-3} (four loops). The truncations at three and at five loops, whose last term in Σ is negative, have no acceptable solution at all for $\lambda \geq 5$ and for $\lambda \geq 2$ respectively (dashes in the table; the entry 1.0000 at $\lambda = 3$ is a solution, but a poor one).
3. **The stationary value is more accurate than the propagator**, as stated in Sec. 1.6.

The applications below use the two-loop truncation.

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2. The CJT action of the two models

2.1 Variables

The fields are the fermion field ψ and the complex Vorton-Scalar Φ_{VS} (S1, Eqs. (1.1) and (3.1)). The variables of the CJT action are:

Variable	Meaning	Its stationarity condition
ϕ	expectation value, $\langle\Phi_{VS}\rangle = \phi/\sqrt{2}$ (real by a choice of phase); the fermion mass is $g_{VS}\phi$	field equation of the Vorton-Scalar in the state considered
D_s, D_ϑ	propagators of the radial and of the phase mode	their gap equations (masses in a medium or at finite temperature)
$\mathcal{S}$	propagator of the doubled fermion field: normal blocks and pair blocks	normal blocks: mass and energy of the fermions; pair blocks: the gap equation

The action is the sum of (1.5) for the two boson modes and (1.10) for the fermions:

$$\begin{aligned}\Gamma[\phi, D_s, D_\vartheta, \mathcal{S}] = S_{\rm cl}[\phi] + \frac{1}{2}\sum_{i=s,\vartheta}\Big\{\mathrm{Tr}\ln D_i^{-1} + \mathrm{Tr}\big[D_{0i}^{-1}(\phi)D_i - 1\big]\Big\} \\ - \frac{1}{2}\mathrm{Tr}\ln\mathcal{S}^{-1} - \frac{1}{2}\mathrm{Tr}\big[\mathcal{S}_0^{-1}(\phi)\,\mathcal{S} - 1\big] + \Gamma_2[\phi, D_s, D_\vartheta, \mathcal{S}].\end{aligned} \tag{2.1}$$

2.2 The two-loop Γ_2

With the vertices of S1, Eq. (1.4), there are three kinds of 2PI diagram with two loops:

- **(a) a closed fermion line with one boson line across it**, for each of the two modes. Its derivative with respect to the fermion propagator is the self-energy of a fermion that emits and reabsorbs a boson: the normal blocks give the exchange (Fock) energy of Sec. 5.4, the pair blocks give the pairing interaction of Secs. 3 and 4 and, without a medium, the ladder equation of Sec. 7. Its derivative with respect to a boson propagator is the fermion loop in the boson self-energy.
- **(b) two boson loops touching at a four-boson vertex** (the "double bubble"), from the quartic terms of the potential of the Vorton-Scalar. It gives the bosons masses that depend on the temperature (Sec. 6).
- **(c) two three-boson vertices joined by three boson lines**, proportional to $\lambda_{VS}^2\phi^2$. It is omitted, as in the Hartree approximation.

No diagram is needed for the mean field of the Vorton-Scalar acting on the fermions: that effect is contained in the dependence of $S_{\rm cl}$ and of $\mathcal{S}_0^{-1}(\phi)$ on ϕ, and it appears when Γ is varied with respect to ϕ (Sec. 5).

2.3 Which part is used where

Sec.	State	Variables kept	Further approximations
3, 4	paired state in a medium	pair blocks of $\mathcal{S}$	static exchange; particles only; ϕ, D_i and the normal blocks at their values without pairing
5	unpaired state in a medium, $T = 0$	ϕ; normal blocks of $\mathcal{S}$	exchange energy to first order; bosons at tree level
6	no medium, temperature T	ϕ, D_s, D_ϑ	fermions at one loop; diagram (b) only
7	vacuum	pair blocks of $\mathcal{S}$, all four momentum components	linearized; no change of the normal blocks

2.4 The symmetry that organizes the results

In the sector used throughout (fields independent of the four extra coordinates) the models have an internal symmetry that was computed for this supplement from the matrices (S1, Sec. 2.1). The complete output of the program:

item (computed from the matrices)	7+1	4+4
dimension of the commutant, momenta along x1 x2 x3	8	8
dimension of the commutant, momenta along x0 x1 x2 x3	4	4
dimension of the commutant, momenta along every space coordinate of the model	1	4
dimension of the centre of the zero-mode commutant	2	2
$\|P^2 - 1\|$ for the traceless central generator P	0	0
dimensions of the two ideals	[3, 3]	[3, 3]
structure constant $\|f_{123}\|$ of each ideal (generators with $\mathrm{Tr}\, T_i T_j = 2\delta_{ij}$)	1.000000000000 1.000000000000	1.000000000000 1.000000000000
$\|\sum_i T_i^2 - \frac{3}{4}$ (projector)$\|$ for each ideal	0 0	2.2×10^{-16} 2.2×10^{-16}
positive-energy states with P = +1 and P = -1 at a test momentum	4.000000 4.000000	4.000000 4.000000
the four vector pair operators $C^{-}\gamma^a$, $a \in \{0, 5, 6, 7\}$, transform among themselves: residual	0	3.1×10^{-16}
$\|$Casimir $- \frac{3}{4}\|$ of each ideal on the four pair operators	0 0	5.5×10^{-17} 5.5×10^{-17}
fermion number of the pair operators	2.000000	2.000000
charge of the pair operators under the central generator P	0	0
number of invariant quadratic forms on the four pair operators	1	1
the invariant form (diagonal; axes 0 5 6 7; first entry 1)	+1.000000 +1.000000 +1.000000 +1.000000	+1.000000 -1.000000 -1.000000 -1.000000
largest off-diagonal entry of the invariant form	0	2.2×10^{-16}
complex dimension of the generators that mix particles with antiparticles, $\mu = 0$	2	2
the same with a chemical potential	0	0
$\gamma^0\gamma^5$	antihermitian; commutator with the dynamics 0	Hermitian; commutator with the dynamics 0
$\gamma^0\gamma^6$	antihermitian; commutator with the dynamics 0	Hermitian; commutator with the dynamics 0
$\gamma^0\gamma^7$	antihermitian; commutator with the dynamics 0	Hermitian; commutator with the dynamics 0

item (computed from the matrices)	7+1	4+4
$\gamma^5\gamma^6$	antihermitian; commutator with the dynamics 0	antihermitian; commutator with the dynamics 0
$\gamma^5\gamma^7$	antihermitian; commutator with the dynamics 0	antihermitian; commutator with the dynamics 0
$\gamma^6\gamma^7$	antihermitian; commutator with the dynamics 0	antihermitian; commutator with the dynamics 0

All 24 items have the expected values (`data/reduced_symmetry.csv`, column `expected`; 24 of 24 flagged as correct by the program).

In words: the symmetry is $SU(2)_A \times SU(2)_B \times U(1)_F \times U(1)_E$ in both spacetimes; the four vector pair operators form a 4-vector of $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$ with fermion number 2; and when the chemical potential vanishes there are four more real generators, which mix particles with antiparticles (the two lines before the last six; Sec. 7.1). In 7+1 the $SO(4)$ consists of rotations of the extra space axes; in 4+4 half of its generators, the Hermitian matrices $\gamma^0\gamma^t$, have no geometric meaning.

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3. The paired state: checks of the results of S1

3.1 The potential

Keep in (2.1) the pair blocks of the fermion propagator and diagram (a), with the three approximations of S1, Sec. 5.1 (static exchange; particles only; the normal state unchanged). A propagator with pair blocks is then specified by a gap function $\Delta(k)$: the quasi-particles have the energies $E_k = \sqrt{\xi_k^2 + \Delta(k)^2}$, $\xi_k = \sqrt{m^2 + k^2} - \mu$, and the pair correlation is

$$\kappa(k) = \langle b(-\vec{k})\, b(\vec{k})\rangle = \frac{\Delta(k)}{2E_k} \tanh \frac{E_k}{2T}. \tag{3.1}$$

The sums over the Matsubara frequencies in (2.1) can be done in closed form. For a unitary state (all $\nu = 8$ internal states have the same gap), and measured from the unpaired state at the same T and μ, the result is

$$\boxed{\begin{aligned} V[\Delta] = \nu \int \frac{d^3k}{(2\pi)^3} \Big\{ \frac{|\xi_k| - E_k}{2} - T \ln \frac{1 + e^{-E_k/T}}{1 + e^{-|\xi_k|/T}} + \Delta(k)\,\kappa(k) \Big\} \\ - \frac{\nu}{2} \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3}\; \kappa(k)\, W(k,k')\, \kappa(k') \end{aligned}} \tag{3.2}$$

with the kernel W of S1, Eq. (5.3). The first line comes from the trace terms of (1.10): the energy and the entropy of the quasi-particles, and the term $\Delta\kappa$ that the trace of $\mathcal{S}_0^{-1}\mathcal{S}$ contributes. The second line is Γ_2: one boson exchanged between two pairs. Because $\partial/\partial\Delta$ of the first two terms in the braces is $-\kappa$,

$$\frac{\delta V}{\delta \Delta(k)} = \nu\, \frac{\partial \kappa}{\partial \Delta} \Big[\Delta(k) - \int \frac{d^3k'}{(2\pi)^3} W(k,k')\, \kappa(k') \Big], \tag{3.3}$$

and $\partial\kappa/\partial\Delta > 0$: **the potential is stationary exactly where the gap equation (5.2) of S1 holds.** The unit test `cjt_pairing::tests::gradient_against_finite_differences` checks (3.3) against numerical derivatives of (3.2).

3.2 Four checks

The program evaluates (3.2) on the momentum grid of the command `gap`, for twelve of its parameter sets: eight in the vector channel, two each in the two-form channel and with the radial mode alone.

Check 1: an independent solution. The gap equation is solved by Newton's method with the exact Jacobian, starting from a function of different height and shape (`Potential::newton`); the command `gap` used fixed-point iteration. The two gap functions agree at every grid point within 1.1×10^{-11} of the gap, and the gaps stored in `data/gap_scan.csv` within 7.9×10^{-9}.

Check 2: the solution is a minimum of the potential. Along the family $s\,\Delta(k)$ of rescaled gap functions the potential has its minimum at $s = 1$ (within 5.1×10^{-7}). All second variations are positive: the matrix of second derivatives is $D^{1/2}(1 - \tilde{K})D^{1/2}$ with a positive diagonal matrix D and $\tilde{K}_{ij} = \sqrt{\omega_i \kappa_i'}\, W_{ij} \sqrt{\omega_j \kappa_j'}$ (ω_i the integration weights), and the smallest eigenvalue of $1 - \tilde{K}$ is positive (column "smallest curvature"; Cholesky factorization and inverse iteration).

Check 3: the value at the minimum is the condensation energy. In weak coupling it must approach $V_{\mathrm{BCS}} = -\nu N(0) \Delta_F^2 / 4$, with $N(0) = k_F E_F' / 2\pi^2$ the density of states per internal state and Δ_F the gap at the Fermi momentum. The ratio lies between 0.987 and 1.000, closest to 1 for the smallest gaps.

Check 4: the critical temperature. Below T_c the unpaired state $\Delta = 0$ is unstable: the second variation of V there has a negative direction. T_c is the temperature at which its determinant changes sign (Gaussian elimination, bisection); it agrees with the T_c of the linearized gap equation (power iteration in `gap.rs`) within 2.6×10^{-12}, the resolution of the bisection.

In the table m_s and k_F are in units of the fermion mass m; "pair" names the pair operator (S1, Sec. 6); the three columns of differences are relative.

g_{VS}	m_s	k_F	pair	gap$/E_F$	Newton vs fixed point	vs `gap_scan.csv`	V/V_{BCS}	smallest curvature	T_c/gap	T_c: two methods
2	0.3	0.5	vector	5.84×10^{-2}	3.3×10^{-12}	2.3×10^{-9}	0.9977	0.257	0.5660	2.6×10^{-12}
2	0.3	1	vector	1.38×10^{-2}	4.8×10^{-12}	3.5×10^{-9}	0.9997	0.201	0.5668	2.6×10^{-12}
2	1	1	vector	5.03×10^{-4}	1.1×10^{-11}	7.4×10^{-9}	1.0000	0.114	0.5669	2.6×10^{-12}
2	0.1	1	vector	3.42×10^{-2}	2.8×10^{-12}	2.2×10^{-9}	0.9936	0.263	0.5641	2.6×10^{-12}
3	1	0.35	vector	3.68×10^{-2}	5.7×10^{-12}	4.8×10^{-9}	0.9993	0.157	0.5666	2.6×10^{-12}
3	1	0.5	vector	5.84×10^{-2}	4.7×10^{-12}	4.1×10^{-9}	0.9981	0.194	0.5660	2.6×10^{-12}
3	1	1	vector	6.86×10^{-2}	4.1×10^{-12}	3.0×10^{-9}	0.9969	0.243	0.5655	2.6×10^{-12}
3	0.3	1.5	vector	1.28×10^{-1}	3.2×10^{-12}	1.7×10^{-9}	0.9873	0.330	0.5634	2.6×10^{-12}
2	0.3	0.5	two-form	2.22×10^{-2}	4.3×10^{-12}	3.1×10^{-9}	0.9997	0.219	0.5668	2.6×10^{-12}
3	1	0.5	two-form	3.20×10^{-4}	1.0×10^{-11}	7.9×10^{-9}	1.0000	0.106	0.5669	2.6×10^{-12}
2	0.3	0.5	radial only	3.72×10^{-2}	3.7×10^{-12}	2.5×10^{-9}	0.9991	0.239	0.5666	2.6×10^{-12}
3	1	0.5	radial only	9.38×10^{-3}	5.9×10^{-12}	4.7×10^{-9}	0.9999	0.154	0.5669	2.6×10^{-12}

3.3 The bound pair, from the gap equation

Without a Fermi sea, at $T = 0$ and $\mu < m$, the linearized gap equation reads $\Delta(k) = \int \frac{d^3k'}{(2\pi)^3} W(k,k')\Delta(k')/[2(E_{k'}-\mu)]$. With $\psi = \Delta/[2(E-\mu)]$ and $\mu = m - E_b/2$ it becomes, for momenta small compared with m,

$$\Big(\frac{k^2}{m} + E_b\Big)\psi(k) = \int \frac{d^3k'}{(2\pi)^3}\, W(k,k')\,\psi(k') :$$

the Schrödinger equation of the relative motion of two particles (reduced mass $m/2$) with the potential $-W$, which for the radial mode is the Yukawa potential (4.6) of S1. **The instability of the vacuum of pairs and the bound state of two particles are the same thing**; this is how the paired state of overlapping pairs goes over into a condensate of bound pairs (S1, Sec. 5.3). A bound state appears when the largest eigenvalue of the kernel at $\mu = m$ reaches 1. The program finds the corresponding $D_{VS} = \alpha_{VS} m/m_s$ with the relativistic kernel:

m_s/m	D_c from the gap equation	relative difference from `dimer.rs`
0.3	1.73473	+3.27e-02
0.1	1.69016	+6.16e-03
0.03	1.68096	+6.88e-04
0.01	1.67994	+8.13e-05
0.003	1.67982	+7.47e-06

For a force of long range compared with the Compton wavelength of the fermions the result is 1.67982; the program `dimer.rs`, which integrates the radial Schrödinger equation in coordinate space, gave 1.67981. The two programs share no code. For m_s comparable to m relativistic effects raise the threshold by a few per cent (first lines of the table).

3.4 The Ginzburg–Landau coefficients

Near T_c the potential can be expanded along a gap function of fixed shape, $\Delta(k) = s\,\varphi(k)$, where φ is the solution of the linearized gap equation normalized to 1 at the Fermi momentum: $V = \alpha(T)\,s^2 + \frac{1}{2}\beta\,s^4 + \ldots$ For a unitary state $\mathrm{Tr}[\mathsf{M}\mathsf{M}^\dagger] = 8s^2$, so that the coefficients of S1, Sec. 6.2, are $a = \alpha/8$ and $b = \beta/8$. Expanding (3.2) and using $\int W\kappa'\varphi = \eta(T)\,\varphi$, where $\eta(T)$ is the largest eigenvalue of the linearized gap equation ($\eta(T_c) = 1$), one finds

$$\alpha(T) = \frac{\nu}{2}[1-\eta(T)]\int \frac{d^3k}{(2\pi)^3}\kappa'\varphi^2, \qquad \frac{\beta}{2} = \nu\Big[\frac{1}{8} - \frac{\eta(T)}{6}\Big]\int \frac{d^3k}{(2\pi)^3}\kappa'''\varphi^4, \tag{3.4}$$

with κ', κ''' the first and third derivatives of (3.1) with respect to Δ at $\Delta = 0$. At T_c the bracket of β is $-1/24$, and with a constant density of states (3.4) gives the formulas of S1. Away from T_c, $\eta(T) - 1 \approx -\lambda \ln(T/T_c)$ with $\lambda = N(0)W$, so that the coefficients of the potential along a fixed ray carry factors $1 + O(\lambda \ln(T/T_c))$ — for β the factor $1 + 4(\eta - 1)$. The comparison is therefore made at T_c (the program fits V/s^2 at six small amplitudes for $T/T_c = 0.99$ and 1.01). The gradient coefficient K is obtained independently, from the dependence of the pair susceptibility on the momentum of the pair (`pair_bubble_curvature`).

g_{VS}	m_s/m	k_F/m	λ	T_c/E_F	a: potential / formula	b: potential / formula	K: susceptibility / formula
3	1	0.5	0.208	3.3×10^{-2}	0.9993	0.9949	1.0101
2	0.3	1	0.205	7.8×10^{-3}	0.9998	0.9996	1.0007

g_{VS}	m_s/m	k_F/m	λ	T_c/E_F	a: potential / formula	b: potential / formula	K: susceptibility / formula
2	1	1	0.115	2.9×10^{-4}	1.0000	1.0003	1.0000

The three coefficients of S1 are confirmed within 1.0 per cent; the deviations decrease with the coupling, as corrections of relative order T_c/E_F and λ should. For the first parameter set the temperature dependence is:

T/T_c	$\alpha/[4N(0)\ln(T/T_c)]$	$\beta/[7\zeta(3)N(0)/(2\pi^2T_c^2)]$
0.95	1.0099	1.1494
0.98	1.0034	1.0530
0.99	1.0013	1.0232
1.01	0.9972	0.9667
1.02	0.9951	0.9399
1.05	0.9891	0.8650

Both ratios pass through 1 at T_c. The third column varies as $(T_c/T)^2[1+4(\eta-1)]$, that is with the logarithmic slope $-(2+4\lambda)=$ -2.83 according to (3.4); the measured slope is -2.84.

---

4. The paired state: further predictions

4.1 Thermodynamics

At the solution of the gap equation, (3.2) is the difference of the thermodynamic potentials of the paired and the unpaired state. Figure 2 and the table show it, together with the gap, the entropy S and the heat capacity $C=T\,dS/dT$, for $g_{VS}=3$, $m_s=m$, $k_F=0.5\,m$ in the vector channel ($T_c/E_F=3.3\times10^{-2}$).

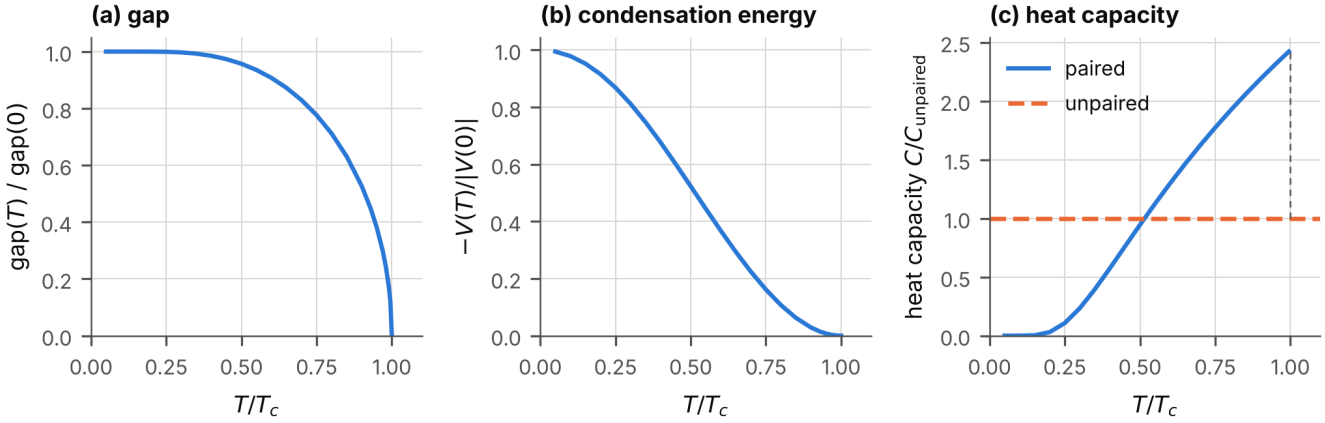


Figure 2: The paired state against temperature. (a) Gap at the Fermi momentum. (b) Condensation energy (minus the CJT potential), in units of its value at $T=0$. (c) Heat capacity in units of that of the unpaired state at the same temperature; it jumps at T_c. Data: `data/cjt_thermo.csv`.

T/T_c	gap(T)/gap(0)	$-V(T)/\lvert V(0)\rvert$	S_s/S_n	C_s/C_n
0.10	1.0000	0.9789	0.0000	0.0000
0.20	0.9999	0.9156	0.0036	0.0304
0.30	0.9971	0.8120	0.0409	0.2340
0.40	0.9851	0.6766	0.1302	0.5743
0.50	0.9570	0.5233	0.2561	0.9443
0.60	0.9072	0.3675	0.4005	1.2973
0.70	0.8291	0.2242	0.5521	1.6217
0.80	0.7114	0.1071	0.7045	1.9170
0.90	0.5266	0.0285	0.8544	2.1863
0.95	0.3806	0.0073	0.9278	2.3123
0.99	0.1731	0.0003	0.9857	2.4093

The universal numbers of weak coupling are reproduced, with corrections of the expected size for a gap of 6 per cent of the Fermi energy:

$$\frac{\Delta(0)}{T_c} = 1.7668\ (1.764), \qquad \frac{C_s - C_n}{C_n}\Big|_{T_c} = 1.433\ (1.426),$$
$$\frac{\Delta(T)}{T_c\sqrt{1 - T/T_c}}\Big|_{0.98T_c} = 3.045\ (3.06 \text{ at } T_c).$$

These hold for both spacetimes and for any unitary state. Two consequences are used elsewhere: the entropy of the paired state is exponentially small below about $T_c/4$ (the paired fluid of Supplement S3 is cold in the strict sense), and the condensation energy per fermion, $\frac{3}{8}\Delta^2/E_F$ in the non-relativistic limit, is a small fraction of the Fermi energy, so that pairing changes the pressure by terms of relative order $(\Delta/E_F)^2$, as assumed in S3, Sec. 4.1.

4.2 The space of the order parameter

The order parameter has four complex components Δ^a along the extra axes $a \in \{0, 5, 6, 7\}$. The program builds the gap matrix `M` of S1, Eq. (6.1), from the Dirac matrices of each spacetime and evaluates the CJT potential at $T = 0$ for a constant attraction in each of the four channels (weak coupling; `ShellModel`), in units in which the gap of the unitary state is 1 and $N(0) = 1$.

The gaps of a general state. In both spacetimes the eight eigenvalues of $\mathtt{M}\mathtt{M}^\dagger$ form two fourfold sets,

$$\boxed{d_\pm^2 = \mathcal{S} \pm \sqrt{\mathcal{S}^2 - \left|\eta_{ab}\Delta^a\Delta^b\right|^2}, \qquad \mathcal{S} = \sum_a |\Delta^a|^2} \tag{4.1}$$

with the metric of the spacetime on the extra axes (unit test `cjt_pairing::tests::order_parameter_space`). The identity (6.2) of S1 is the statement about $d_+^4 + d_-^4$ that follows from (4.1); (4.1) is the complete statement. A state is unitary ($d_+ = d_-$) when $|\eta_{ab}\Delta^a\Delta^b| = \mathcal{S}$.

	state	the two gaps^2	V	second derivatives	flat directions
7+1	hidden-axis singlet	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6647)	Im Δ^0 Re Δ^5 Re Δ^6 Re Δ^7
7+1	along x5	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6647)	Re Δ^0 Im Δ^5 Re Δ^6 Re Δ^7

	state	the two gaps[2]	V	second derivatives	flat directions
7+1	unitary, rotated between x0 and x5	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6648)	(combinations)
7+1	unitary, general direction and phase	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6648)	(combinations)
7+1	half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$; larger gap 1	0.000, 1.000	-1.0000	stationary; unstable	—
7+1	the phases of the other signature's unitary state	0.040, 1.960	-0.9184	not stationary	—
4+4	hidden-axis singlet	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6647)	Im Δ^0 Im Δ^5 Im Δ^6 Im Δ^7
4+4	along x5	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6647)	Im Δ^0 Im Δ^5 Re Δ^6 Re Δ^7
4+4	unitary, rotated between x0 and x5	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6648)	(combinations)
4+4	unitary, general direction and phase	1.000, 1.000	-2.0000	4 flat, 4 positive (6.6648)	(combinations)
4+4	half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$; larger gap 1	0.000, 1.000	-1.0000	stationary; unstable	—
4+4	the phases of the other signature's unitary state	0.040, 1.960	-0.9184	not stationary	—

Reading the table:

1. **The unitary states are the minima**, with $V = -2$, that is $-\nu N(0)\Delta^2/4$, whatever their direction: the hidden-axis singlet, a state along $x5$, and rotated states have the same energy to the accuracy of the calculation, in both spacetimes.
2. **Four flat directions, four massive ones.** Around every unitary state four of the eight real directions cost no energy: the phase (fermion number) and three rotations of the direction (the breaking $SO(4) \to SO(3)$). They are the directions of the four gapless collective modes. The table shows which they are at the hidden-axis singlet: in 7+1 the imaginary part of Δ^0 and the *real* parts of $\Delta^5, \Delta^6, \Delta^7$; in 4+4 the imaginary part of Δ^0 and the *imaginary* parts of the other three — the hidden symmetry rotates Δ^0 into $i\Delta^t$. The other four directions (the amplitude, and three that make the state non-unitary) have equal curvature, 6.6647 $N(0)$.
3. **A half-paired stationary state.** If $\eta_{ab}\Delta^a\Delta^b = 0$, four of the eight internal states are paired with the full gap and four are not paired at all. The state is stationary and has half the condensation energy; it is unstable against opening the second gap (the potential rises by 6.7×10^{-4} when the order parameter is multiplied by 1.02 or by 0.98, and falls by 2.1×10^{-2} when 0.05 is added to Δ^0). It is the analogue of the non-unitary phases of superfluid ^{3}He, which are stabilized by a magnetic field; here the corresponding field would be a chemical potential for one of the generators of $SO(4)$.
4. The configuration that is unitary in one spacetime is far from unitary in the other (last line of each block): the relative phase between Δ^0 and Δ^5 that minimizes the energy is 0 in 7+1 and $\pi/2$ in 4+4.

The curvatures of the potential in the massive directions depend on how propagators away from the stationary point are parametrized; they are not frequencies of collective modes. The number of flat directions and the energies of stationary states do not depend on that choice.

Consequences. Both models have, besides the sound mode, three gapless modes in which the direction of the order parameter in the space of the extra axes oscillates. In 7+1 they are exact Goldstone modes of the model. In 4+4 they are protected by the hidden symmetry only; any dependence of the fields on $x0$ gives them a mass (S1, Sec. 6.3).

5. The unpaired state in a medium: what the Vorton-Scalar does to the fermion mass

5.1 The field equation in matter

Vary (2.1) with respect to ϕ for a Fermi sea without pairing at $T = 0$. The fermion mass is $m^* = g_{VS}\phi$; write $x = m^*/m$. With the tree-level potential of the Vorton-Scalar and the energy of fermions of mass m^* filled up to the Fermi momentum k_F (density $n = \nu k_F^3/6\pi^2$), the energy density in units of m^4 is

$$\varepsilon(k_F, x) = \frac{(x^2-1)^2}{8C} + \nu \int_0^{k_F} \frac{k^2 dk}{2\pi^2} \sqrt{k^2 + x^2} \; [\, + V_{\text{sea}}(x)], \qquad \boxed{C = \frac{g_{VS}^2 m^2}{m_s^2} = \frac{g_{VS}^4}{2\lambda_{VS}} = 4\pi D_{VS} \frac{m}{m_s}} \tag{5.1}$$

and stationarity in x is the field equation of the Vorton-Scalar (S1, Eq. (1.8)) with the scalar density of the fermions as source:

$$x^2 = 1 - 2C\,\nu \int_0^{k_F} \frac{k^2 dk}{2\pi^2} \frac{1}{\sqrt{k^2 + x^2}} \quad \text{or} \quad x = 0. \tag{5.2}$$

Everything depends on the single number C. At low density (5.2) gives $m^* = m\,(1 - C\,n/m^3)$: the mass falls linearly with the density. The term in square brackets in (5.1) is the energy of the filled negative-energy states, which changes when the mass changes. It is divergent and must be renormalized; the symmetry $U(1)_{VS}$ allows two counterterms, which are fixed by requiring that the vacuum value and the mass of the radial mode keep their tree-level values:

$$V_{\text{sea}}(x) = -\frac{\nu}{32\pi^2}\left[x^4 \ln x^2 - \frac{3}{2}x^4 + 2x^2 - \frac{1}{2}\right], \qquad V_{\text{sea}}(1) = V'_{\text{sea}}(1) = V''_{\text{sea}}(1) = 0. \tag{5.3}$$

Results are given without and with this term; the difference measures how much depends on the quantum fluctuations of the fermions.

5.2 Results: the abnormal state, and self-bound matter

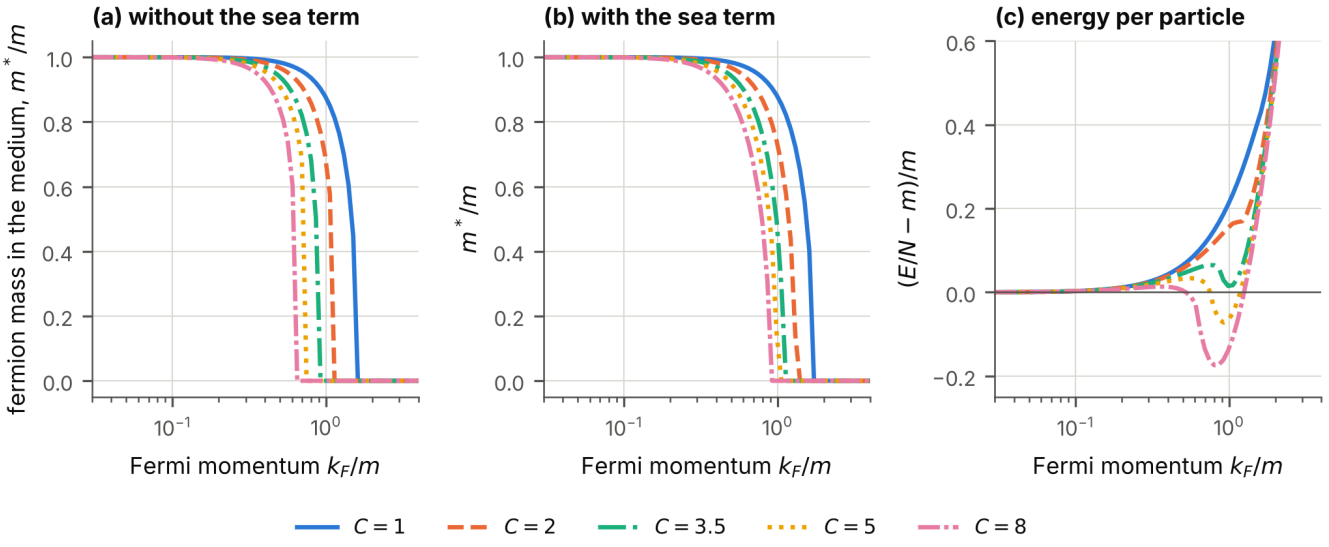


Figure 3: The unpaired state at zero temperature. (a), (b) Fermion mass in the medium against the Fermi momentum, without and with the energy of the negative-energy states. (c) Energy per particle relative to the mass of a free fermion; a minimum below zero means that matter is self-bound. Data: `data/cjt_normal.csv`.

C	$E/N-m$	k_F^{sat}	k_F^{abn}	m^*/m before	k_F^{inst}	with sea: $E/N-m$	k_F^{abn}	k_F^{inst}
0.5	not bound	—	2.226	0.10	—	not bound	2.336	—
1	not bound	—	1.576	0.17	—	not bound	1.724	—
2	not bound	—	1.123	0.35	1.023	not bound	1.317	—
3	not bound	—	0.932	0.43	0.750	not bound	1.151	1.135
3.5	not bound	—	0.867	0.48	0.676	not bound	1.100	0.923
4	-0.019	0.981	0.818	0.50	0.589	not bound	1.059	0.776
5	-0.072	0.928	0.742	0.53	0.479	not bound	0.999	0.610
6	-0.114	0.886	0.688	0.55	0.403	-0.002	0.955	0.479
8	-0.175	0.825	0.610	0.59	0.305	-0.043	0.901	0.351
12	-0.255	0.745	0.516	0.64	0.209	-0.091	0.840	0.224
20	-0.344	0.656	0.423	0.68	0.124	-0.135	0.790	0.129
35.5	-0.432	0.568	0.340	0.71	0.072	-0.168	0.755	0.072

Columns: $E/N-m$ lowest energy per particle of matter at any density (in units of m; "not bound" if it is not below zero); k_F^{sat} Fermi momentum of that state; k_F^{abn} Fermi momentum above which the stable state has $m^*=0$, and m^*/m just below it; k_F^{inst} Fermi momentum above which the pressure falls with increasing density. Momenta in units of m.

1. **Above a critical density the Vorton-Scalar has no expectation value and the fermions are massless.** This is the "abnormal state" of matter that Lee and Wick predicted for any theory in which fermions obtain their mass from a scalar field. Without the sea term the mass jumps to zero (the jump is small for small C); with it the mass falls continuously, to the resolution of the scan. For weak coupling the transition is at $k_F=\pi m/(2\sqrt{C})$.
2. **Matter is self-bound if C exceeds a critical value**:

$$C>\frac{3\pi^2}{8}=3.70\quad\text{(without sea term)},\qquad C>\frac{3\pi^2}{5}=5.92\quad\text{(with it)}.$$

   The program finds 3.7011 and 5.9218. The closed values follow from the energy of the abnormal state: massless fermions with pressure equal to the energy density B of the state without expectation value, as in the bag model of hadrons, have the energy per particle $(24\pi^2B/\nu)^{1/4}$, and $B=m^4/8C$ at tree level, $B=m^4/8C+\nu m^4/64\pi^2$ with (5.3). The bound state is always the abnormal one. For large C a fermion in matter weighs much less than a free one: 43 per cent less for $C=35.5$ without the sea term, 17 per cent with it.
3. **The dilute gas is mechanically unstable above a low density** (k_F^{inst}). At low density the criterion is $k_F>\pi^2m/4C$, or, in terms of the scattering length $a=-g_{VS}^2m/4\pi m_s^2$ of the mean-field interaction, $k_F|a|>\pi/16$.

5.3 A warning about strong coupling

The same function (5.3) shows that for field values *larger* than the vacuum value the one-loop potential of the Vorton-Scalar, $(x^2-1)^2/8C+V_{\text{sea}}(x)$, eventually decreases without bound: the negative term $x^4\ln x^2$ wins (for weak coupling when $\ln x^2>\frac{3}{2}+4\pi^2/\nu C$). The program locates the turning points (`vacuum_instability`; `data/cjt_vacuum_stability.csv`):

C	maximum of the potential at $\phi/v_{VS}=$	below the vacuum beyond $\phi/v_{VS}=$
0.1	8.6×10^{10}	1.1×10^{11}
0.5	229	294
1	19.3	24.7

C	maximum of the potential at $\phi/v_{VS}=$	below the vacuum beyond $\phi/v_{VS}=$
2	5.33	6.74
3.7011	2.75	3.39
5.9218	1.98	2.36
10	1.54	1.77
20	1.26	1.37
35.5	1.14	1.21

For $C \ll 1$, the regime of the weakly coupled superfluid, the instability lies at exponentially large fields and is irrelevant. But **the region of C in which matter is self-bound is a region in which the model, taken at one loop, has no stable vacuum**: for $C \geq 3.7$ the potential falls below its vacuum value at fields only 1.2 to 3.4 times the vacuum value (last column). There the model needs a completion (a larger self-coupling, or new physics at that scale). This is the analogue of the instability that a heavy fermion induces in the potential of the Higgs field.

5.4 The exchange energy

The normal blocks of diagram (a) add the exchange (Fock) energy. To first order (with the propagators of Sec. 5.1 and static exchange) it is

$$E_x = \frac{g_{VS}^2}{2}\,\frac{\nu}{2}\int\!\!\int_{k,k'<k_F} \frac{d^3k\,d^3k'}{(2\pi)^6}\,\frac{EE' \pm m^{*2} - \vec{k}\cdot\vec{k}'}{EE'}\,\frac{1}{|\vec{k}-\vec{k}'|^2 + M^2},$$

upper sign and $M = m_s$ for the radial mode, lower sign and $M = 0$ for the phase mode (the factor in the middle is the trace of the two vertices between the projectors on particle states). Both are positive. In the limit of a heavy radial mode the first reduces to $(g_{VS}^2/2m_s^2)(n^2+\rho_s^2)/2\nu$, the Fierz rearrangement of the scalar interaction (unit test `fock_energy_contact_limit`).

C	m_s/m	g_{VS}^2	k_F/m	m^*/m	radial mode (%)	phase mode (%)
2	0.3	0.18	0.1	1.000	+11.1	+0.3
2	0.3	0.18	0.3	0.993	+6.5	+0.3
2	0.3	0.18	0.6	0.943	+3.1	+0.3
2	0.3	0.18	1	0.679	+1.2	+0.3
2	1	2.00	0.1	1.000	+12.4	+3.1
2	1	2.00	0.3	0.993	+11.5	+3.1
2	1	2.00	0.6	0.943	+9.4	+3.2
2	1	2.00	1	0.679	+5.7	+3.8
2	3	18.00	0.1	1.000	+12.5	+28.1
2	3	18.00	0.3	0.993	+12.6	+28.2
2	3	18.00	0.6	0.943	+12.5	+29.1
2	3	18.00	1	0.679	+10.3	+34.1
5	1	5.00	0.1	0.999	+12.4	+3.1
5	1	5.00	0.3	0.982	+11.4	+3.2
5	1	5.00	0.6	0.832	+8.4	+3.6
5	1	5.00	1	0.000	+2.8	+6.0

The last two columns give the exchange energies in per cent of the mean-field energy shift. The exchange of the radial mode reduces the attraction by at most 12.6 per cent — the factor $1/\nu$ expected for $\nu = 8$ internal states — and less when the range of the force exceeds the distance

between the particles. The exchange of the phase mode is of relative size $2m_s^2/\nu^2 m^2$ at low density: negligible for a light radial mode, but larger than the radial term for $m_s = 3m$. The thresholds of Sec. 5.2 are therefore accurate to about ten per cent for $m_s \lesssim m$, and the mean-field criterion for the instability becomes $k_F|a| > \pi/14$ when exchange is included (factor $1 - 1/\nu$).

5.5 Consequences for the dark-matter analysis of S3

Sound speed of the degenerate gas. S3 used the sound speed of free fermions. The mean field changes it by a factor $1 - O(k_F|a|)$. Collisions of clusters of galaxies limit the scattering cross section per mass to about 1 cm^2/g; with $\sigma = 4\pi a^2$ this gives $k_F|a| < 6.4 \times 10^{-13}$ for fermions of 1 keV in a dwarf galaxy. The correction is negligible, and the gas is far from the instability of Sec. 5.2.

Bound clusters. For the heavy, strongly coupled particles that S3 found to be acceptable cold dark matter ($\alpha_{VS} = 1$, $m \approx 200$ MeV, $m_s \approx 120$ MeV), $C = 35.5$: large clusters are self-bound and in the abnormal state, with 43 per cent of the rest mass of the fermions radiated away as binding energy (17 per cent with the sea term). Whether they form is a question of the history of the early universe that is not addressed here; the warning of Sec. 5.3 applies to this regime.

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6. Finite temperature: the fermion mass melts

Without a medium of fermions but at a temperature T of the dark sector, vary (2.1) with respect to ϕ and to the two boson propagators, with diagram (b) and the fermions at one loop (`cjt_normal.rs`, `Thermal`). The potential is

$$V = U_{\text{eff}}(\phi) + \sum_{i=s,\vartheta} \Big[Q(M_i) + \frac{1}{2}\big(a_i(\phi) - M_i^2\big) P(M_i) \Big] + \frac{\lambda_{VS}}{4}\big[3P_s^2 + 3P_\vartheta^2 + 2P_s P_\vartheta\big],$$

where U_{eff} is the tree-level potential plus the fermion loop (its vacuum part (5.3) and its thermal part), $a_s = U''_{\text{eff}}$, $a_\vartheta = U'_{\text{eff}}/\phi$, $P(M)$ is the thermal part of the boson loop and $Q(M)$ the thermal part of $\frac{1}{2}\mathrm{Tr}\ln$. The last term is diagram (b). Stationarity in the masses gives the gap equations $M_s^2 = a_s + 3\lambda_{VS}P_s + \lambda_{VS}P_\vartheta$ and $M_\vartheta^2 = a_\vartheta + \lambda_{VS}P_s + 3\lambda_{VS}P_\vartheta$.

Two closed estimates of the temperature at which the expectation value vanishes follow from the curvature of the potential at $\phi = 0$:

$$T_{\text{high}}^2 = \frac{12\,[\lambda_{VS}v_{VS}^2 + \nu g_{VS}^2 m^2/8\pi^2]}{4\lambda_{VS} + \nu g_{VS}^2}, \qquad T_N^2 = \frac{12\,[\lambda_{VS}v_{VS}^2 + \nu g_{VS}^2 m^2/8\pi^2]}{2\lambda_{VS} + \nu g_{VS}^2}. \tag{6.1}$$

The first is the one-loop result at high temperature; the second replaces the boson loops by their form for a large number of components, in which the phase mode stays massless. The program follows the state of lowest potential and determines the temperature $T_{1/2}$ at which the expectation value has fallen to half its vacuum value.

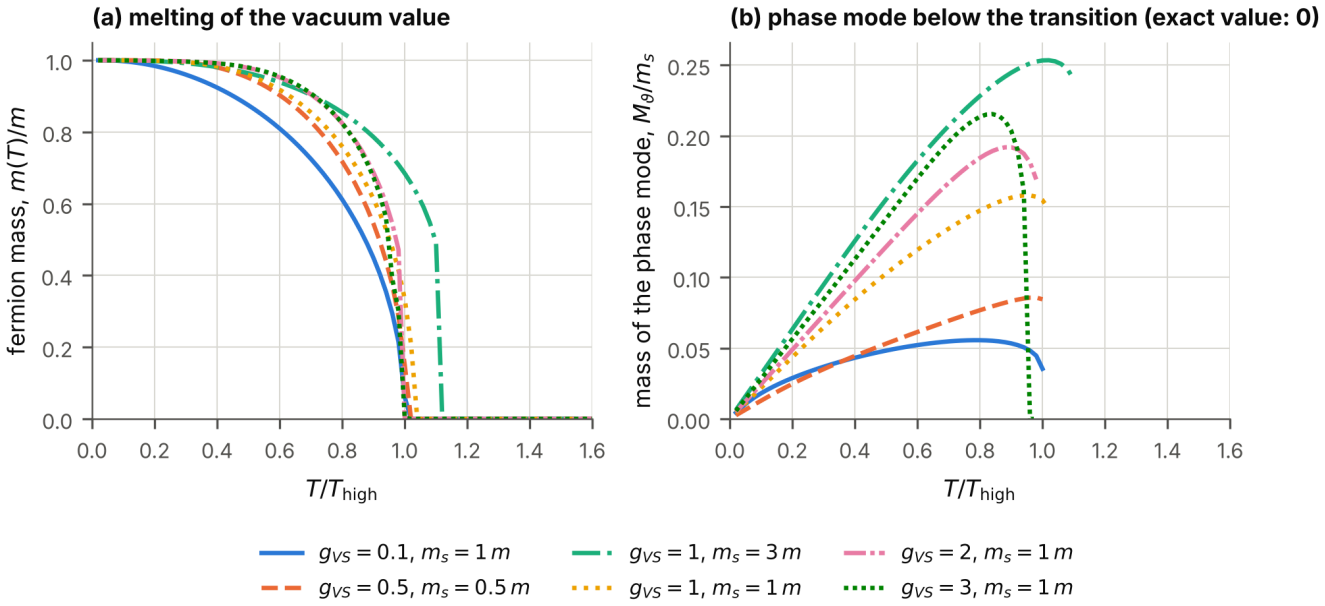


Figure 4: Finite temperature. (a) The expectation value of the Vorton-Scalar, that is the fermion mass in units of its vacuum value, in the state of lowest potential. (b) The mass that the Hartree approximation gives to the phase mode below the transition; in the exact theory it vanishes. Data: `data/cjt_thermal_profile.csv`.

g_{VS}	m_s/m	C	T_{high}/v	T_N/v	$T_{1/2}/v$	T_0/v	ϕ/v before T_0	$T_{1/2}/m$
0.1	1	0.01	0.775	0.817	0.677	0.776	0.07	6.77
0.3	1	0.09	0.782	0.824	0.698	0.788	0.08	2.33
0.5	0.5	1	0.461	0.468	0.423	0.464	0.08	0.85
1	3	0.111	1.457	1.802	1.599	1.632	0.41	1.60
1	1	1	0.849	0.895	0.808	0.875	0.10	0.81
2	1	4	1.042	1.099	1.014	1.042	0.37	0.51
3	1	9	1.302	1.372	1.238	1.302	0.14	0.41

Columns: $T_{1/2}$ temperature at which the expectation value has fallen to half; T_0 temperature at which it has fallen to zero (below 2 per cent of v_{VS}, the resolution of the scan); ϕ/v at $0.998\,T_0$.

1. **The fermion mass melts** at a temperature of order v_{VS}: $T_{1/2}$ agrees with the two estimates (6.1) within 17 per cent. In units of the fermion mass it ranges from 0.4 (strong Yukawa coupling) to 7 (weak).
2. **The order of the transition is not reliably predicted.** For the sets $(g_{VS}, m_s/m) = (1, 3)$, $(2, 1)$ the expectation value jumps to zero from about 0.4 v_{VS}; for the others it falls steeply but, as far as the scan can tell, continuously (a square-root law, $\phi/v_{VS} = \sqrt{1 - T^2/T_0^2}$, gives 0.06 at $0.998\,T_0$; the table has 0.07 to 0.14). The Hartree approximation is known to give a discontinuous transition for a pure scalar theory, where the true transition is continuous; a jump found here is therefore not evidence that the transition of the model is discontinuous. The table records what the truncated equations give.
3. **The defect of the approximation is visible** in panel (b): below the transition the phase mode has a mass of up to 0.25 m_s where Goldstone's theorem requires zero (Sec. 1.6).

Consequence for the Hubble-tension analysis of S3. There the fermions annihilate when the temperature falls below about $m/3$, at constant mass. This is justified if the mass is already at its vacuum value then, $T_{1/2} \gg m/3$, which holds for $g_{VS} \lesssim 0.3$ (first two lines of the table). For $g_{VS} \gtrsim 1$

the fermions acquire their mass at a temperature below m and annihilate at once: the step of the dark radiation would coincide with the transition. That case was not simulated; since the simulated steps are already excluded, it would not change the verdict unless the transition released the entropy much earlier than the simulated range, in which case the early-step family of S3 applies.

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7. The vacuum: can the pair condense without a medium?

7.1 The symmetry without a chemical potential

When $\mu = 0$ the sector of fields independent of the extra coordinates has more symmetry than in a medium: four further real generators exist, which mix the field with its conjugate (computed in `symmetry.rs`; Sec. 2.4). The reason is that the 16 components are equivalent to four ordinary Dirac fields, or eight Majorana fields, with equal masses; the radial mode couples to all eight alike, the phase mode with opposite signs to the four with $P = +1$ and the four with $P = -1$. The symmetry is $O(4) \times O(4)$, of dimension 12. The pair condensate of fermion number 2 is one orientation of a 16-component real multiplet that also contains particle–antiparticle condensates; a chemical potential reduces the symmetry to the $U(2) \times U(2)$ of Sec. 2.4 and singles out the pairs. Whatever its orientation, the condensate does the same thing to the spectrum: it splits the masses of two combinations of fields into $m + \sigma$ and $m - \sigma$.

7.2 The ladder equation

The stationarity condition of (2.1) for the fermion propagator with diagram (a), in four-dimensional Euclidean momentum space, is

$$2\sigma(p) = g_{VS}^2 \int \frac{d^4k}{(2\pi)^4} \left[f(m+\sigma) - f(m-\sigma) \right] \Big[\frac{1}{(p-k)^2 + m_s^2} + \frac{1}{(p-k)^2} \Big], \qquad f(M) = \frac{M}{k^2 + M^2}. \tag{7.1}$$

Both modes contribute with the same sign in this channel, as in the medium. This is the "ladder" (or "rainbow") approximation that CJT themselves used for dynamical symmetry breaking. Linearized in σ, after the angular integration, it is an eigenvalue problem with the kernel $\frac{g_{VS}^2}{8\pi^2}\, k^3 \frac{k^2 - m^2}{(k^2+m^2)^2}\, [A(p,k;m_s) + A(p,k;0)]$, $A = 2/[p^2 + k^2 + M^2 + \sqrt{(p^2+k^2+M^2)^2 - 4p^2k^2}]$, and a cutoff Λ where the four-dimensional description ends. A condensate forms when the largest eigenvalue exceeds 1.

Test. For massless fermions, one massless boson and momenta between μ and Λ the equation is solved by $\sigma \propto p^{-1} \cos(\rho \ln p^2 + \delta)$, and the smallest critical coupling is $g^2/16\pi^2 = \frac{1}{4} + \rho^2$ with $\rho \ln(\Lambda/\mu) = \arctan(1/2\rho)$. The program reproduces this within 0.2 per cent (unit test `cjt_vacuum::tests::scale_invariant_ladder_equation`). With two massless bosons and $\Lambda \to \infty$ the critical coupling is $g^2/8\pi^2 = \frac{1}{4}$, that is $\alpha_c = \pi/2$.

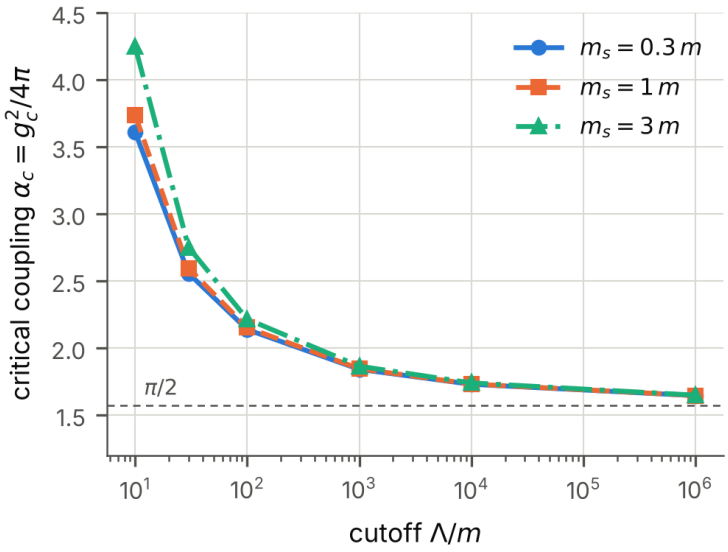


Figure 5: Critical coupling for a pair condensate in the vacuum, against the cutoff. Data: `data/cjt_vacuum.csv`.

m_s/m	$\Lambda/m = 10^1$	$\Lambda/m = 30$	$\Lambda/m = 10^2$	$\Lambda/m = 10^3$	$\Lambda/m = 10^4$	$\Lambda/m = 10^6$
0.3	3.609	2.556	2.137	1.840	1.728	1.644
1	3.737	2.593	2.152	1.844	1.730	1.645
3	4.246	2.749	2.214	1.863	1.739	1.647

A pair condensate in the vacuum requires $\alpha_{VS} > \alpha_c$, between 1.64 and about 4 depending on the cutoff. For comparison: two fermions bind for $D_{VS} = \alpha_{VS} m/m_s > 1.68$, so at such couplings bound pairs exist unless $m_s \gtrsim m$; and by Sec. 5.3 the one-loop potential of the Vorton-Scalar is then unstable unless its self-coupling is equally strong. The ladder approximation neglects the corrections to the vertices and to the boson propagators, which are not small at these couplings. The result is a statement about where to look, not a proof that a condensate forms.

Consequence for the Higgs hypothesis (S2). The condition "the condensate exists in empty space" can be met only in a strongly coupled Vorton-Scalar sector. This replaces a cruder estimate in an earlier draft of S2 (error log E28).

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8. What has been established, and what has not

Established:

1. The numerical results of S1 for the paired state — gap functions, critical temperatures, condensation energy, the critical coupling of the bound pair, and the three Ginzburg–Landau coefficients — are reproduced by a different formulation (the CJT potential) and different algorithms, to the accuracies of the table in Sec. 0.
2. The paired solution is a minimum of the potential; the unpaired state is unstable below T_c.
3. The spectrum of gaps of a general state, Eq. (4.1); the manifold of ground states; four flat directions; the half-paired stationary state — for both spacetimes, from their matrices.
4. Within the mean-field treatment: the fermion mass in matter, the abnormal state, the thresholds $3\pi^2/8$ and $3\pi^2/5$ for self-bound matter, and the size of the exchange corrections.
5. Within the Hartree approximation: the melting temperature of the fermion mass, to about 20 per cent.

6. Within the ladder approximation: the critical coupling for a vacuum condensate.

Not established:

1. Everything beyond the two-loop truncation of Γ_2, and beyond the further approximations listed in Sec. 2.3. In particular the frequency dependence of the pairing interaction (the phase mode is massless, and its exchange is not instantaneous), which is known to change the prefactor of the gap in comparable systems.
2. The order of the thermal transition (Sec. 6), and all properties of the strong-coupling regime (Secs. 5.3 and 7).
3. The frequencies and dampings of the collective modes; only their number and the static energies were determined.
4. The masses that the three additional gapless modes of 4+4 acquire when the fields depend on $x0$.
5. Paired states in the abnormal phase, and the pairing of massless fermions.

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9. Provenance of the numbers and figures

Item	Part of `spacetimes cjt`	Source file	Data file
internal symmetry (Sec. 2.4)	command `symmetry`	`symmetry.rs`	`data/reduced_symmetry.csv`
zero-dimensional test (Sec. 1.7, Fig. 1)	`toy`	`cjt_toy.rs`	`data/cjt_toy.csv`
checks of the paired state (Sec. 3.2)	`pairing`	`cjt_pairing.rs`	`data/cjt_pairing_check.csv`
threshold of the bound pair (Sec. 3.3)	`threshold`	`cjt_pairing.rs`	`data/cjt_threshold.csv`
Ginzburg–Landau coefficients (Sec. 3.4)	`thermo`	`cjt_pairing.rs`	`data/cjt_gl.csv`, `cjt_gl_summary.csv`
thermodynamics (Sec. 4.1, Fig. 2)	`thermo`	`cjt_pairing.rs`	`data/cjt_thermo.csv`
space of the order parameter (Sec. 4.2)	`landscape`	`cjt_pairing.rs`	`data/cjt_landscape.csv`
unpaired state in matter (Sec. 5, Fig. 3)	`normal`	`cjt_normal.rs`	`data/cjt_normal.csv`, `cjt_normal_summary.csv`, `cjt_fock.csv`, `cjt_vacuum_stability.csv`
finite temperature (Sec. 6, Fig. 4)	`thermal`	`cjt_normal.rs`	`data/cjt_thermal.csv`, `cjt_thermal_profile.csv`
vacuum (Sec. 7, Fig. 5)	`vacuum`	`cjt_vacuum.rs`	`data/cjt_vacuum.csv`
tables and single numbers of this document	—	`code/scripts/make_tables_cjt.py`	`provenance/generated/tables.md`
figures	—	`code/scripts/make_figures.py`	`figures/fig_cjt_*.pdf`, `.png`

The unit tests that guard each part are named in the text; all are run by `cargo test --release`.

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